

Missing Data Imputation in a Diabetes Dataset

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1. Introduction

This report presents a machine-learning-based approach for handling missing data in a clinical diabetes dataset containing 1,487 records. Since missing values were present in eight of the nine variables, each incomplete column was treated as a separate prediction problem. Multiple candidate models were evaluated for every variable, and the best-performing model was used to impute the missing values. The quality of the imputation was then assessed through both prediction performance and its impact on a downstream diabetes classification task.

2. Dataset Description

The working dataset comprises 1,487 patient records across nine attributes spanning demographic, clinical-history, and laboratory measurements:

- Demographic: gender, age
- Clinical history (binary): hypertension, heart_disease, smoking_history (multi-class)
- Laboratory: bmi, HbA1c_level, blood_glucose_level
- Target: diabetes (binary)

Of the 1,487 records, 450 (30.3%) are complete, whereas 1,037 (69.7%) contain at least one missing value. Missing data are present in all eight predictor variables, with the number of missing entries ranging from 64 in gender to 249 in heart_disease.

2.1 Missingness Profile

Missing data were present in several predictor variables, although the extent of missingness varied across columns. The proportion of missing values ranged from approximately 4.3% in the gender variable to 16.7% in heart_disease, while the target variable contained no missing observations.

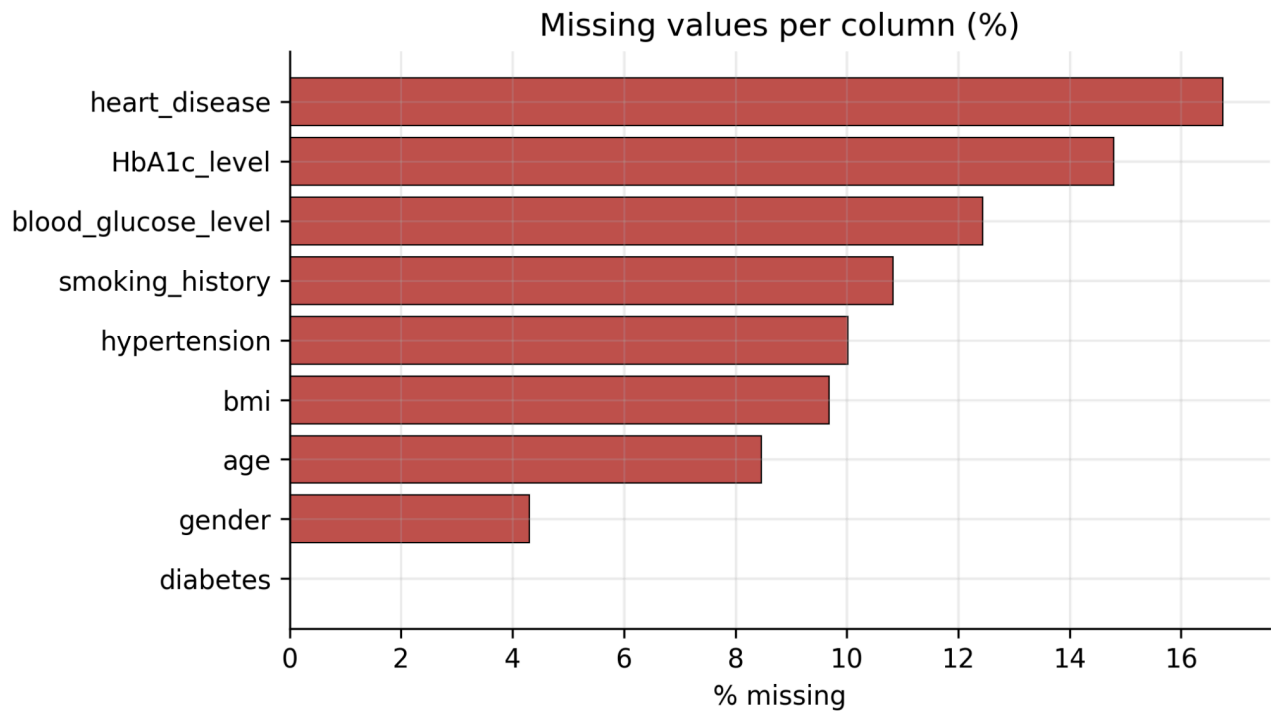


Figure 1. Percentage of missing values by column.

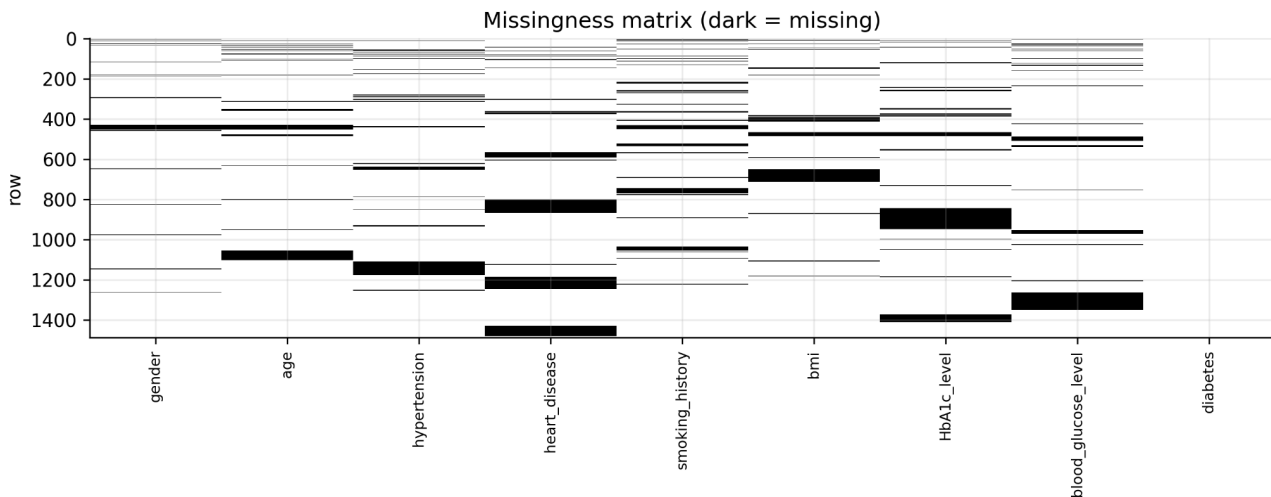


Figure 2. Missingness matrix (dark bands indicate a missing cell)

The missingness matrix shows that missing values are distributed across different variables and records, without any obvious concentration in a specific subset of observations. No strong overlapping pattern is visible among the columns, suggesting that the missing data are spread throughout the dataset rather than being confined to a particular group of records. This distribution supports the use of column-wise predictive imputation, where each incomplete variable is estimated using information from the remaining observed attributes.

2.2 Univariate Distributions

Before imputation, the numeric variables show the expected clinical shapes: a right-skewed bmi concentrated near 25–30, a bimodal age distribution with a spike near 80, and HbA1c_level / blood_glucose_level each showing a secondary high-value mode consistent with a diabetic subpopulation.

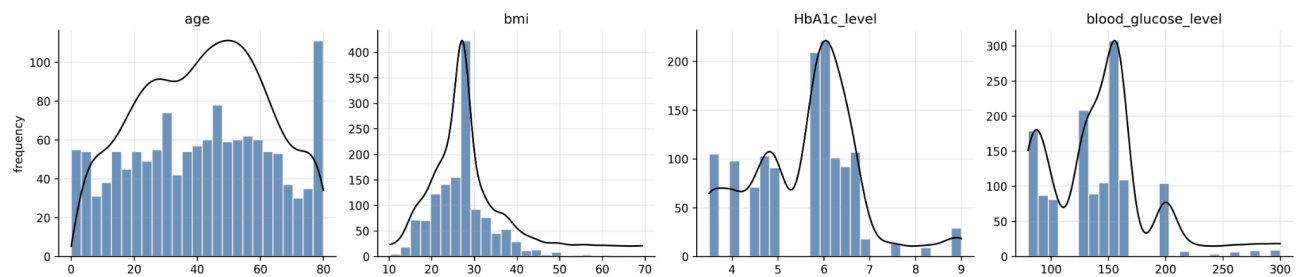


Figure 3. Numeric distributions (observed values)

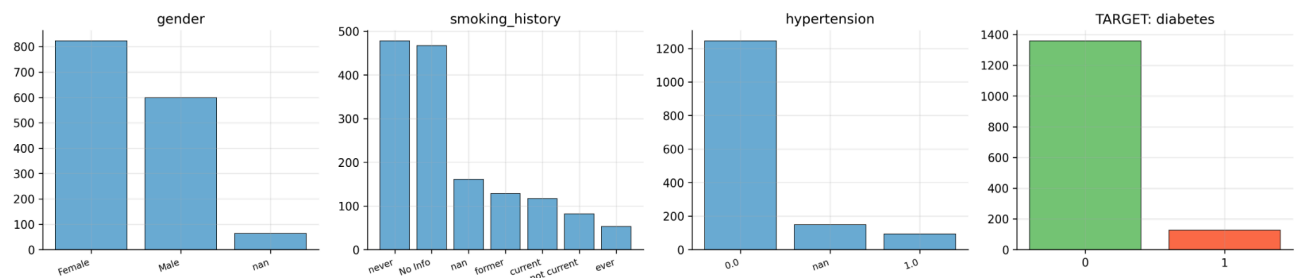
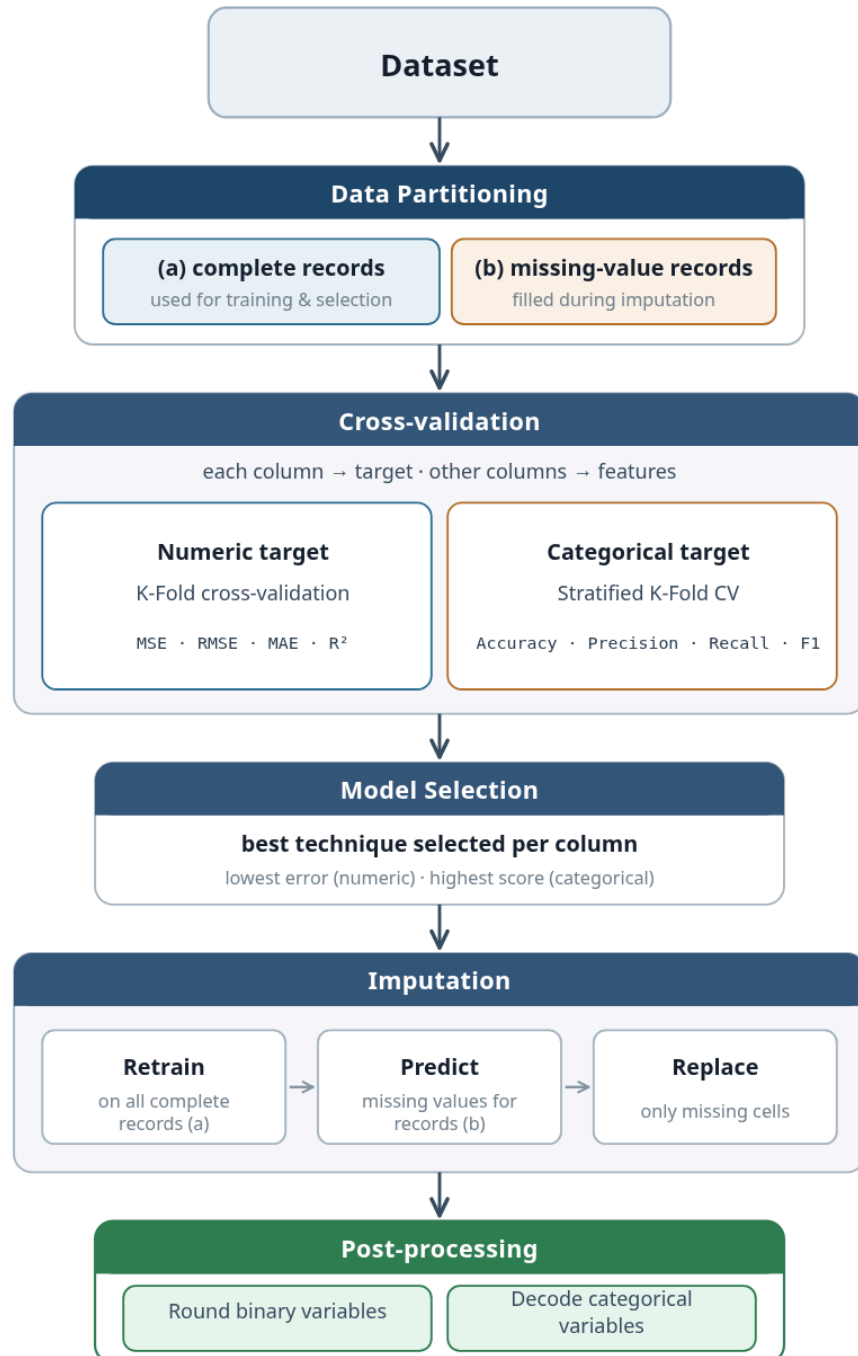


Figure 4. Categorical distributions and target imbalance

3. Methodology

Imputation is treated as a column-wise supervised learning task rather than a single global rule. The overall pipeline is structured as follows:



1. The dataset is divided into two subsets: (a) complete records and (b) records containing missing values.
2. Using only the complete records, each column is treated sequentially as the target variable while all other columns are used as predictor variables.
3. For numerical target columns, fifteen imputation techniques are evaluated using K-Fold Cross-Validation, and their performance is compared using metrics such as MSE, RMSE, MAE, and R^2 .
4. For categorical target columns, fifteen imputation techniques are evaluated using Stratified K-Fold Cross-Validation, and their performance is compared using metrics such as Accuracy, Precision, Recall, and F1-score.
5. For each column, the best-performing imputation technique is selected based on the evaluation results—choosing the method with the lowest error for numerical targets and the highest performance for categorical targets.
6. The selected model is retrained using all available complete records for the corresponding column.
7. The trained model is used to predict missing values in that column.
8. Only the originally missing entries are replaced with predicted values, while observed values remain unchanged.
9. Finally, post-processing is applied to the imputed dataset, such as rounding binary variables to valid class labels and decoding categorical variables back to their original form.

4. Results

4.1 Numeric Columns

For each numeric column, candidate techniques are ranked by cross-validated RMSE (lower is better). The spread between the best and worst technique varies substantially by column: `bmi` and `HbA1c_level` are nearly flat across all candidates (RMSE ≈ 6.2 – 7.0 and ≈ 1.0 – 1.2 respectively), implying limited predictive headroom beyond the leading model, whereas `blood_glucose_level` and `age` show a wider gap, with NeuralNet consistently the weakest performer across all four columns.

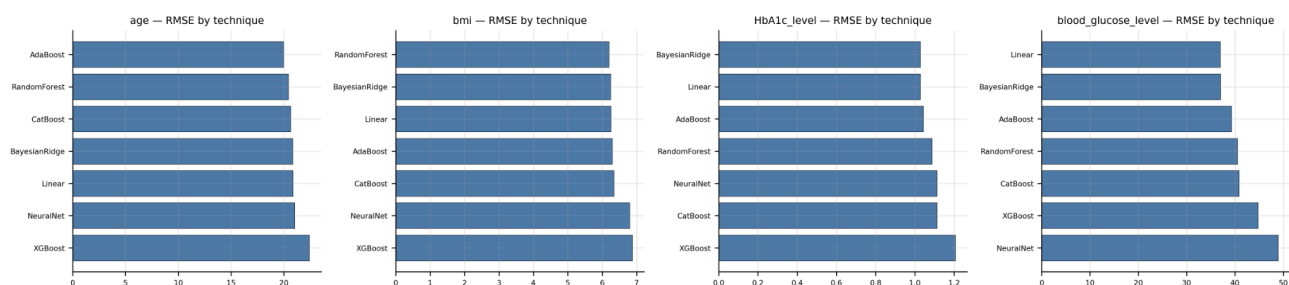


Figure 6. Cross-validated RMSE by technique for each numeric column requiring imputation (lower is better).

Selected technique per numeric column:

Column	Selected Technique	MSE	RMSE	MAE	R ²
age	AdaBoost	398.51	19.96	16.23	0.270
bmi	RandomForest	38.43	6.20	4.62	0.137
HbA1c_level	BayesianRidge	1.06	1.03	0.88	0.178
blood_glucose_level	Linear	1366.24	36.96	30.25	0.102

AdaBoost was selected for age and Random Forest for bmi, both ensemble methods that can pick up mild non-linear interactions among the demographic and clinical-history features; HbA1c_level favoured the regularized linear BayesianRidge, and blood_glucose_level was best captured by an ordinary linear model. All four selected models post a positive R² (0.10–0.27), confirming each column carries genuine, exploitable signal in the remaining features rather than being effectively unpredictable — though the R² values are modest in absolute terms, consistent with these labs being driven substantially by physiology not captured elsewhere in the table.

blood_glucose_level still posts the largest absolute error (RMSE $\approx$ 37) of the four numeric columns, reflecting this variable's wide dynamic range (roughly 80–300) rather than a methodological failure; HbA1c_level, by contrast, has the smallest absolute error but also the narrowest dynamic range.

4.2 Categorical Columns

For categorical columns, macro-F1 is used to rank techniques because it weights minority and majority classes equally, exposing models that simply predict the majority class for an unfair F1 advantage.

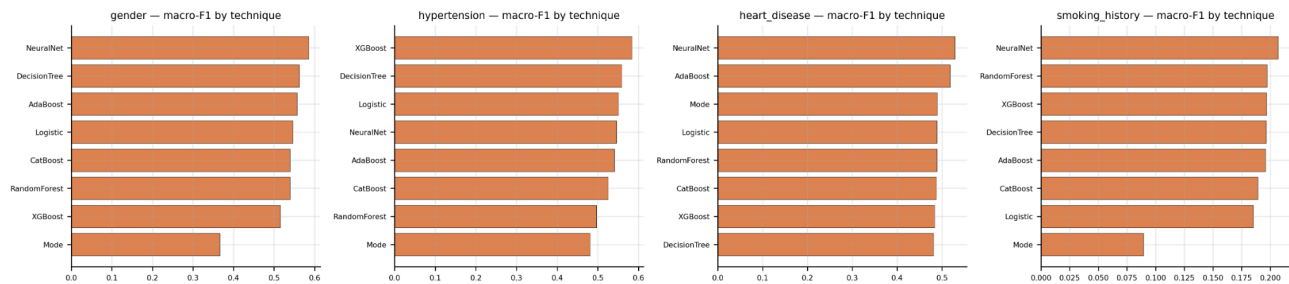


Figure 7. Cross-validated macro-F1 by technique for each categorical column requiring imputation (higher is better).

The Mode baseline is included for comparison; the gap between Mode and the best technique is itself informative, e.g. it is large for hypertension/heart_disease (relative imbalance is exploitable) and small for smoking_history (limited recoverable signal).

Selected technique per categorical column:

Column	Selected Technique	Accuracy	Precision	Recall	Macro-F1
gender	NeuralNet	0.613	0.601	0.588	0.585
hypertension	XGBoost	0.909	0.614	0.576	0.584
heart_disease	NeuralNet	0.958	0.531	0.531	0.529
smoking_history	NeuralNet	0.476	0.195	0.228	0.207

heart_disease and hypertension both achieve very high raw accuracy (0.96 and 0.91), but this is substantially inflated by class imbalance the macro-F1 values (0.53 and 0.58) show that minority-class recall is far from solved. gender lands in the middle (macro-F1 = 0.585), and smoking_history, a seven-level categorical variable, was again the hardest to recover (macro-F1 = 0.207), unsurprising given the variable encodes a behavioural / self-reported history with comparatively weak ties to the clinical and demographic features used as predictors.

4.3 Distributional Integrity After Imputation

To verify the imputation did not distort the underlying variable distributions, the post-imputation density of each numeric column was overlaid against the pre-imputation (observed-only) density.

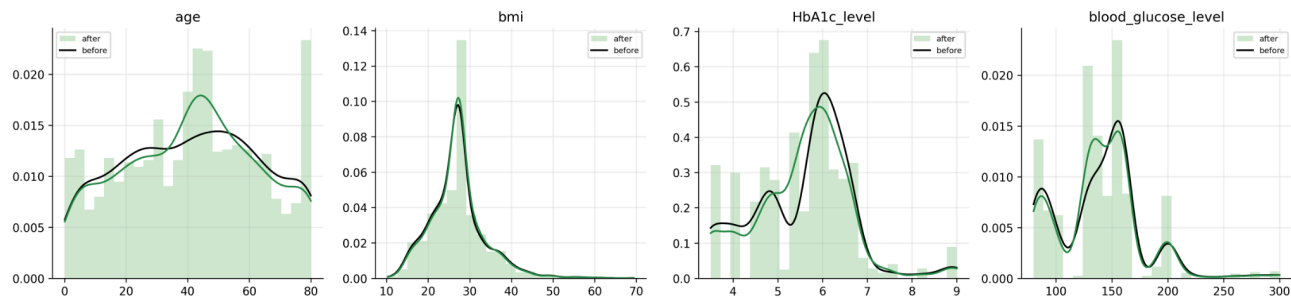


Figure 8. Pre- and post-imputation density comparison for numeric variables

Overall, the post-imputation distributions preserve the key features of the original data (peak location, skew direction, secondary modes), which is the qualitative check we want before trusting the filled values in correlation or downstream modelling analysis.

5. Downstream Validation: Diabetes Classification

Imputation quality metrics (e.g., RMSE and F1-score) evaluate how accurately individual variables are reconstructed. However, the most important question is whether the imputed dataset remains suitable for the downstream predictive task. To assess this, a diabetes classification model was trained on the fully imputed dataset and evaluated on a held-out test split.

5.1 Classifier Leaderboard

Model	RMSE	Accuracy	Precision	Recall	F1	ROC AUC
Random Forest	0.133	0.980	0.955	0.808	0.875	0.992
XGBoost	0.161	0.966	0.864	0.731	0.792	0.981
CatBoost	0.121	0.983	1.000	0.808	0.894	0.995

CatBoost is the strongest of the three, reaching 98.3% accuracy with perfect precision, 80.8% recall on the positive (diabetic) class, and an AUC of 0.995. Random Forest is close behind; XGBoost trails on recall and F1, missing more diabetic cases than the other two. All three substantially outperform a majority-class baseline, given diabetes prevalence of roughly 9% in this cohort.

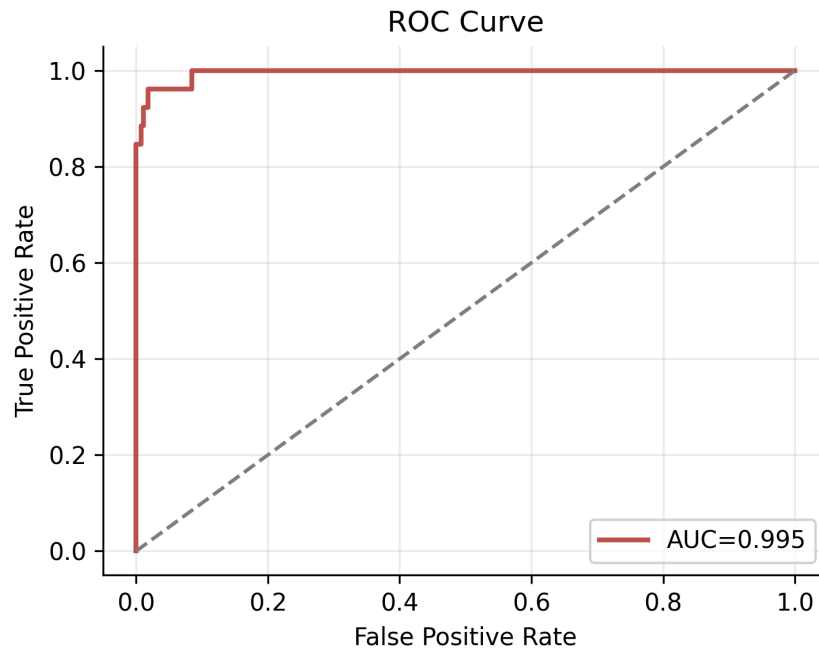


Figure 11. ROC curve (best classifier, AUC = 0.995)

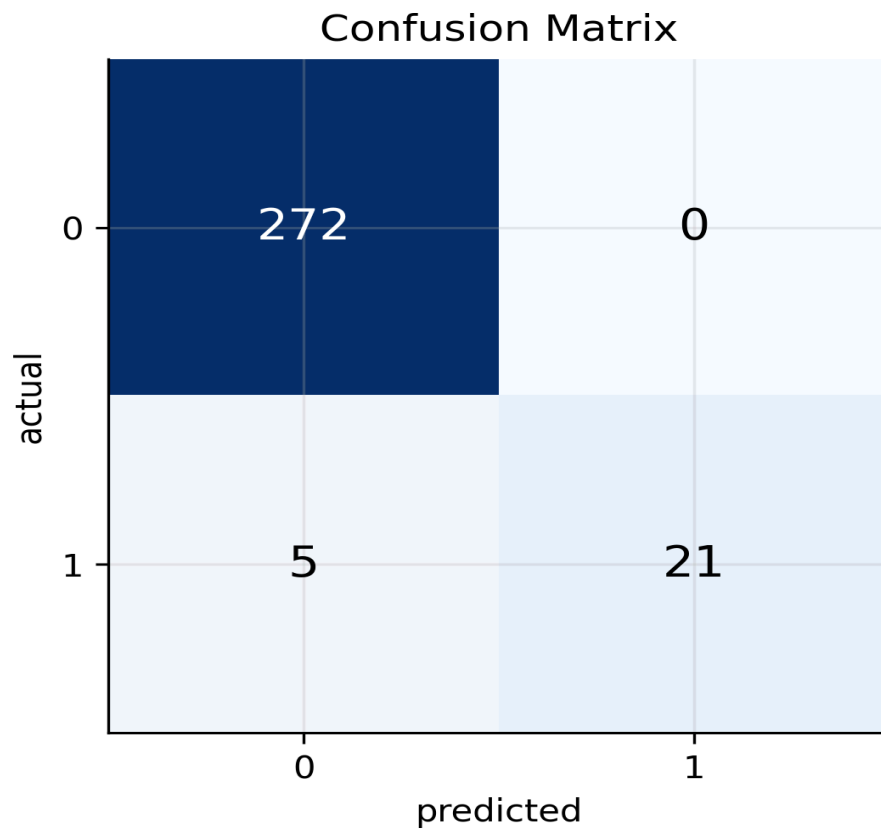


Figure 12. Confusion matrix (best classifier, test set results)

These results should be interpreted with two caveats in mind. First, although the test set contains $n = 298$ samples, the positive class remains a minority (26 diabetic cases). As a result, recall is more sensitive to individual misclassifications than for the negative class. While the metrics are encouraging and based on a substantially larger evaluation sample than a 40-row fold would provide, minority-class performance still requires careful interpretation.

5.2 Imputation Quality Re-Examined Against True Values

As an additional check, 20% of known cells in the complete records were masked and re-imputed using the column's selected method, then compared against their true (held-out) values. This cell-wise masking test corroborates the leaderboard rankings in Section 5: hypertension and heart_disease recover with reasonable raw accuracy but modest F1, gender and smoking_history remain the weakest links, and all four numeric columns retain a small positive R^2 .

Column	Type	RMSE / Accuracy	MAE / Precision	R^2 / Recall	F1
age	numeric	21.86	17.70	0.027	—
bmi	numeric	7.44	5.26	0.067	—
HbA1c_level	numeric	1.11	0.86	0.025	—
blood_glucose_level	numeric	40.44	29.81	-0.002	—
gender	categorical	0.622	0.599	0.530	0.472
smoking_history	categorical	0.433	0.202	0.212	0.204
hypertension	categorical	0.956	0.478	0.500	0.489
heart_disease	categorical	0.956	0.483	0.494	0.489

6. Discussion

No single imputation technique is universally optimal.

The best-performing models varied across columns, with AdaBoost, Random Forest, Bayesian Ridge, Linear Regression, XGBoost, and Neural Network (MLP) each selected at least once. This variability strongly supports a per-column model selection strategy, rather than applying a single global imputation method (e.g., KNN or a fixed MICE estimator) across a heterogeneous dataset.

Numeric variables contain recoverable but uneven signal.

All four numeric columns achieved positive cross-validated R^2 values (0.10–0.27) for their selected models. Age showed the strongest predictability, while blood_glucose_level was the weakest. This pattern is consistent with age being influenced by demographic and lifestyle-related features available in the dataset, whereas blood glucose is driven by more acute physiological factors not fully captured by the remaining variables.

7. Comparison with HyperImpute: Generalized Iterative Imputation with Automatic Model Selection

This report employs a per-column model selection strategy applied to a single clinical dataset. To contextualize these results and positioning within the broader imputation literature, this section provides an objective comparison with HyperImpute, a published generalized iterative imputation framework (Jarrett et al., 2022, ICML).

7.1 Methodological Comparison

Core Approach

Our method treats each incomplete column as an independent supervised learning problem, conducting cross-validation on complete records and selecting the best-performing model per column. This is a single-pass approach: models are fit once on complete data and applied directly to impute missing cells.

HyperImpute, by contrast, employs iterative refinement. It begins with baseline imputation (mean substitution) and then iteratively re-evaluates and potentially re-selects models for each column across multiple rounds. The algorithm continues until convergence (typically 4 iterations), with model selection informed by the continuously updated imputation quality across columns.

Model Selection Strategy

Our approach evaluates a fixed pool of models (Linear, Ridge, KNN, SVM, Decision Tree, Random Forest, AdaBoost, XGBoost, CatBoost, Neural Network, plus baselines) using their default hyperparameters. The best model per column is selected based on cross-validated RMSE (numeric) or macro-F1 (categorical).

HyperImpute integrates automatic hyperparameter optimization using an adapted Hyperband algorithm. Rather than using default hyperparameters, each column's selected model has its hyperparameters tuned via a configuration search. This means the best

model for a column also includes the best hyperparameters for that model class, rather than generic defaults.

7.2 Comparative Evaluation

Scope of Validation

Our evaluation is:

- Single dataset (clinical diabetes, $n = 1,487$)
- 8 incomplete variables (4.3–16.7% missingness)
- Includes distributional checks (KDE overlap)
- Correlation stability analysis
- Downstream task validation (diabetes prediction)

HyperImpute evaluation is significantly broader:

- 12 UCI datasets
- MCAR, MAR, MNAR missingness mechanisms
- Multiple missingness ratios (10%–70%)
- Benchmarked against 9 state-of-the-art methods
- Includes sensitivity + ablation studies

7.3 Key Differences

Aspect	Our Approach	HyperImpute
Iteration Strategy	Single-pass	Iterative (typically 4 rounds)
Hyperparameter Tuning	None (defaults only)	Automatic via Hyperband
Optimization	Per-column imputation quality	Per-column quality + cross-column adaptation
Evaluation Scope	One domain, one task	12 datasets, 3 missingness patterns
Status	Technical Report	ICML 2022 (peer-reviewed)
Theoretical Analysis	Empirical only	Formal MAR support, complexity analysis
Generalization	single dataset	Proven across 12 datasets

Appendix: Supplementary Figures

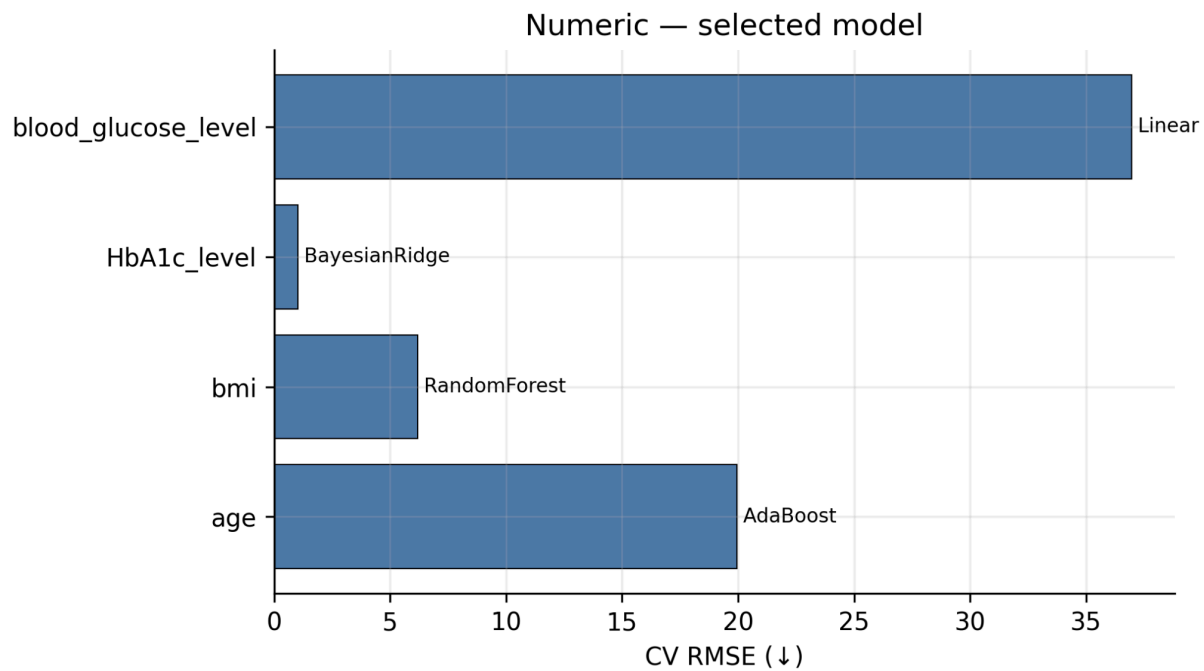


Figure A1. CV RMSE of the selected (winning) technique for each numeric column, summary view.

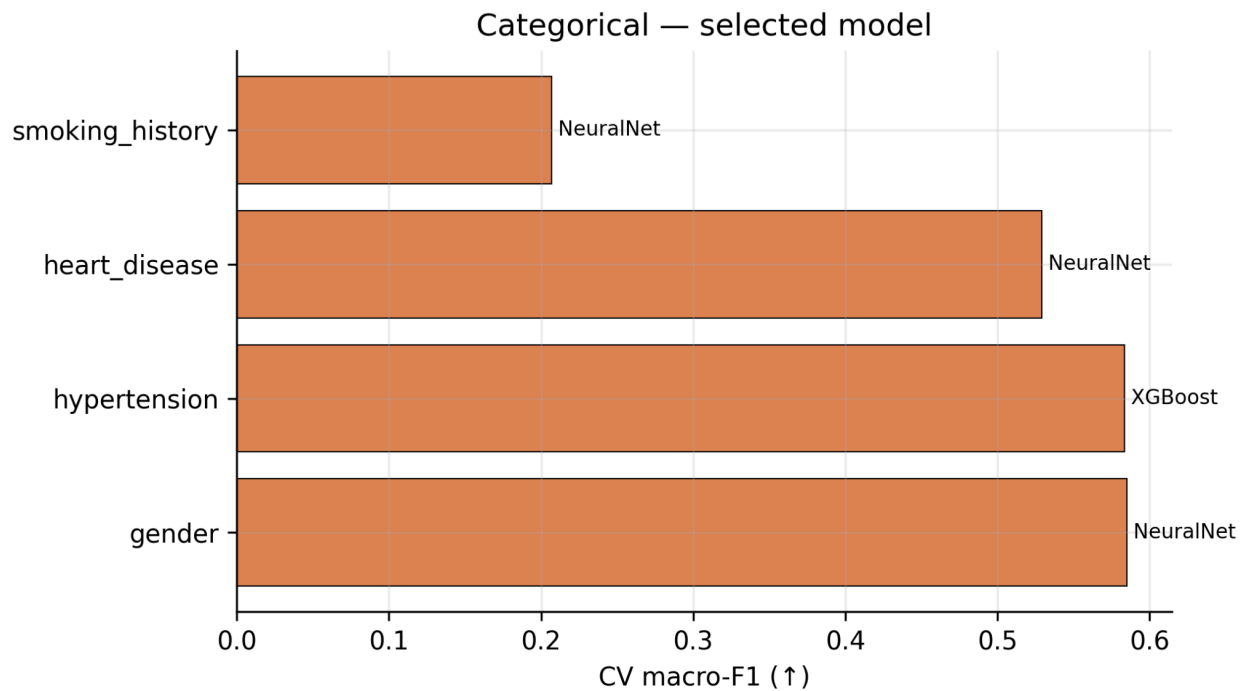


Figure A2. CV macro-F1 of the selected (winning) technique for each categorical column, summary view.